

Chapter 8: Philosophical Implications, Fruitful Applications:

Some Topics for Essays:

1. (From exercise) "As far as the mathematical theorems refer to algebra, they are not sure, and as far as they are sure, they do not refer to reality."

- Albert Fin (1921)

The scale balances by rocks to the blind. A scale demonstrates addition, subtraction, multiplication, and division. Algebra is advantageous during the relationship between total rocks and pounds (lbs.). Mathematical theorems signify and aid other real tools.

2.

Conventionalism:

"Conventionalism" remains in science.

Condensed matter is a division in physics with large attention by universities, professors, textbooks, and societies. However,

temperatures near absolute zero are uncommon. The reality does not support endeavors by scientists.

"Conventionalism" collapsed in sub-atomic physics, certainly in the 'Tevatron.'

Ambition by scientists had an expensive budget, evident by shutdowns. Then, small colliders were advantageous to materials and medicine. Decisions toward reality better matched current science.

3.

Cite:

Blank, A.

"The Luneburg Theory of Binocular Visual Space."

Columbia University, College of Physicians and Surgeons (1953)

Albert classified vision through binoculars by hyperbolic geometry. Main features in the paper were experiments on vision by people.

Critical evidences supported "Luneburg Theory" including, distance between objects, curvature from local points and parallel lights. The publication supported literature for physicians, ophthalmologists, and optometry.

4.

(from exercise) "It can be said that the discovery of Non-Euclidean geometry led to the extensive modern development of mathematical logic. Elaborate..."

Cite:

DeLong, D.

"Lost knowledge"

Oxford University Press (2004)

The citation from Exercise#4 supported corporate retention. Knowledge became lost in both corporation and mathematics. Activities did not assess performance since Descarte and Poincare. Innovation was inefficient

because low retention, routine,
and feedback.

5. (From exercise) Jacques Hadamard said, "Practical application: is found by not looking for it, and one can say that the whole progress of civilization rests on that principle... It seldom happens that important mathematical researches are directly undertaken in view of a given practical use: they are inspired by the desire to know and understand."

David Hilbert maintained, "In spite of the importance of the applications of mathematics, these must never be made the measure of its value.

Jacobi, "The glory of the human spirit is the sole aim of all science."

Lobachevsky believed, "There is no branch of mathematics, however

abstract, that may not someday be applied to the phenomena of the real world."

Comment on these views.

Lobachevsky spoke truth. The Seattle Public Library had a section for mathematics. To a librarian, math applied to law, medicine, education, language, history, technology, astronomy, and statistics. Lobachevsky was correct after two-hundred years. In addition the library also had differential equations, abstract geometry, and elementary geometry.

6.

a) Cite:

Rényi, A.

"Dialogues on Mathematics"

Hungarian Academy of Sciences (1967)

Hippocrates criticized Socrates about mathematics in a narrative. Conversation.

Were comical and playwright.

Hippocrates challenged public opinion, census, and eventually behavior.

Stories about Greek heroes led

Hippocrates into false conclusions.

Mathematicians were trustworthy to

Socrates. In the narrative, Socrates

cultured Hippocrates out from paradox

and into reality.

b) (from exercise) "How do you explain that, as often happens, mathematicians living far from each other and having no contact independently discover the same truths."

Comment on the thought:

Mathematicians progressed logic.

Regions across seas came to similar conclusion because formal relationships.

Math derived from people by a robust system over 1000+ years. Knowledge in both algebra and geometry aided.

mathematicians toward similar topics.

Foundational logic and common events helped mathematicians independently discover the same truths.

7.

(From exercise) Statement by Michael Polanyi:

"We can now turn to the paradox of mathematics.. It reminds one of the clown who solemnly sets up in the middle of the arena two gateposts with a securely locked gate between them.... and carefully locks it after himself - while all the while the whole arena lies open on either side of the gateposts where he could go round them unhindered."

Comment about Polanyi:

The crowd and clown mutually consist in truth. The crowd: simple truth; the clown: complex truth. In the analogy above by Polanyi, both perspectives agreed.

Hopefully abstract thoughts compound into useful tools, rather than philosophical questions.

8.

(from exercise) Hadamard, 1945

"There is a scientific taste just as there is a literary or artistic one... The Greek geometers thought when they investigated the ellipse, because there is no other conceivable way."

Polyani, 1964

"We dwell on mathematics and affirm its statements for the sake of intellectual beauty. For if this passion were extinct, ... its conceptions would dissolve and its proofs carry no conviction."

Comments to Hadamard and Polyani:

What if, similes and hyperboles were logical. Together, Hadamard and Polyani would be accurate. And the precision by the author; Greensburg; metonymy.

9.

(from exercise) Heinrich Hertz

"One cannot escape the feeling that these

mathematical formulas have an independent existence and an intelligence of their own, that they are wiser than we are, wiser even than their own discoverers, that we get more out of them than was originally put into them."

In summary, Heinrich Hertz assigned awareness and abstract thought, broadly intelligence to formula.

10. (from exercise) Kurt Gödel

"Geometrical intuition, strictly speaking, is not mathematical, but rather a priori intuition.

In its purely mathematical aspect our Euclidean ~~space~~ intuition is perfectly correct, namely, it represents correctly a certain structure existing in the realm of mathematical objects. Even physically it is correct "in the small." >>

Response to Gödel:

Kurt's comment generalizes many subjects.

An ad lib without "geometrical, and mathematical" matches topics in Science, history, government, etc. Kurt was correct.

11. (from exercise) Rolf R. Loehrich

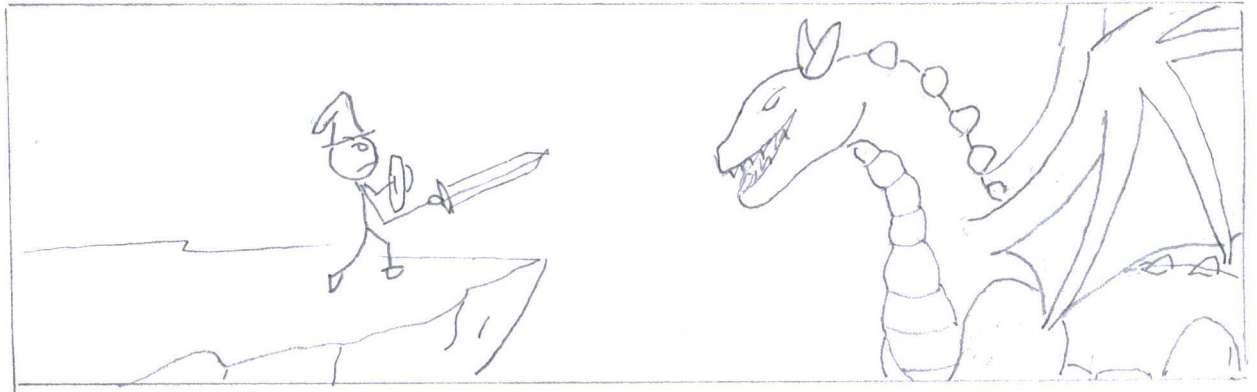
A mathematical system is hardly ever presented axiomatized at its inception. Successful axiomatization is a fruition of an *exercitium cogitandi*. Once a system is axiomatized, mathematical activity can be played as a game, as a manipulation of symbols by virtue of rule-systems thought of as invented."

Response to Loehrich:

Rolf had a clear perspective about mathematics. Symbols built a large

rule-system. 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 combined into the number system

Numbers established elementary operations. While algebra, geometry, and logic separated into subjects for axioms.



12. (from exercise) An essay on the development of geometry in ancient Greece... be particularly interested in the female mathematician Hypatia.

Essay:

Western philosophy classified Hypatia into Roman history, after Greek philosophy and mathematics. Daughter of Theon II, Hypatia saw massive architecture: basilica, arches, cupola,

colosseum, amphitheaters, vaults; and civil infrastructure; aqueducts, bath-houses; common to Rome. In Alexandria, Hypatia built instruments for class. Her inventions measured density, stars, and vaporization. Theon II, her father transcribed 'Euclid's Elements.' She proposed math about conics. Aftermath in Alexandria led to hostility and eventual death.

13. (from exercise) A. Well (1950)
"If logic is the hygiene of mathematicians, it is not his source of food."

Felix Klein

"... Mathematics has been most advanced by those who distinguish themselves by intuition rather than by rigorous proofs"

E. T. Bell (1945)

"Discovery, after all, is more important in science than strict deductive proof. Without discovery there is nothing for deduction to attack and reduce to order."
Compare and contrast:

The three routes are similar. Because math is initial, and discovery after by necessity. An Arabic quote as an analogy, "If we always listened to the leader, then we'd learn no more."

14.

Cite:

Lakatos, I.

"Proofs and Refutations: The logic of mathematical Discovery."

University of Cambridge (1976)

An evil genius is an antagonist to discovery by infallible logic. Euclid created a subject with many fans. The rigid criticisms about 'Euclid's

Elements' promoted Euclid to evil genius.

15.

Cite:

Moise, E.

"Elementary geometry from an advanced standpoint."

Queens College of the City University of New York (1998).

Mr. Moise proposed area in Chapter 24: Hyperbolic Geometry. First parallelograms could have area. The diagonals form triangles with equivalent area in a parallelogram, because similar bases and angles. Moise conjectured, "Hyperbolic triangles have defected-angles less than 180° , but exact areas. Saccheri quadrilaterals are hyperbolic and derive from two triangles ($< 180^\circ$). And so, hyperbolic shapes

have area.

16. Cite:

Russell, B.

"The foundations of Geometry."

New York University (1956)

Russell put hyperbolic geometry into three periods by mathematicians, publications, and arguments for a dissertation in philosophy. The short history delved into epistemology, empirical nature, and intuition. A primary section focused on German Immanuel Kant (1724-1804) because public criticism about geometry. No a priori evidence led Kant to dislike hyperbolic geometry. The dissertation by Russell debated public opinion through personal essays, and other mathematicians.

17.

Cite: Torretti, R.

"Philosophy of Geometry from
Reimann to Poincare."

Puerto Rico (1978)

Reimann's versus Helmholtz's axioms were important to the author. A space coordinated by dimension, distance, and continuity. Helmholtz demonstrated alternative axioms, initially an n -space. Later, points translated and rotated. Torretti cited publications in-detail.

18. (from exercise) Poincare's invented Universe:

a) At any point P inside S , the absolute temperature T is directly proportional to $R^2 - r^2$, where r is the distance from P to the center S .

- b) The length, width, and height of an object vary directly with the absolute temperature of the object.
- c) All objects in U instantaneously take on the temperatures of their locations.
- d) Light travels along the shortest path from one point to another.
- (from exercise) "An inhabitant of U could not detect his change in temperature and size as he moves around with temperature or tape, and that he could never reach boundary S ."

Challenge against Poincare:

Since $T \propto (R^2 - r^2) = \text{constant}$, the temperature never raises or lowers, in addition to size because $(\text{length, width, height}) \propto T$.
The interior S limits size near the core.

The Poincare universe is unusual.

19. (From Exercise) Albert Einstein stated,

"The concept 'true' does not tally with the assertions of pure geometry, because by the word 'true' we are eventually in the habit of designating always the correspondence with a real object; however, geometry is not concerned with the relation of the ideas involving in it to objects of experience, but only with the logical connection of these ideas among themselves."

He believed, "geometry is not a
contingency table."